Defined Water Kefir Fermentations

Water Kefir Isolates



Bifidobacterium +	Yeast +	LAB +	AAB +	*Zymomonas*
Bifidobacterium fermentum	*Brettanomyces bruxellensis*	*Lacticasei-bacillus paracasei*	*Acetobacter lovaniensis*	*Zymomonas mobilis*
	Hanseniaspora valbyensis	*Lentilacto-bacillus hilgardii*	*Acetobacter orientalis*	
Bifidobacterium aquikefiricola	*Pichia fermentans*	*Leuconostoc pseudo-mesenteroides*	*Gluconobacter japonicus*	none
	Saccharomyces cerevisiae	*Liquorilacto-bacillus satsumensis*	*Gluconobacter oxydans*	

Screening



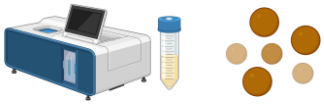
Fermentation setup:
256 defined (= 2 *Bifidobacterium* x 4 yeast x 4 LAB x 4 AAB x 2 *Zymomonas* groups)
+ 12 WK grains + 12 media control; no replicates; 20 ml fermentations
Analysis: aroma (day 4); pH (day 0 and day 3); cells per ml (day 0 and day 3)
Analysis (selected subset n = 72): targeted metabolite analysis

In-depth Analysis



Fermentation setup: 5 defined + 1 WK grain + 1 media control; two biological replicates;
2 l fermentations
Analysis: taxonomic composition (metagenomics); hedonic evaluation;
Optimised Descriptive Profile; targeted metabolite analysis;
untargeted volatile organic compound analysis

Influence of AAB



Fermentation setup: 4 defined + 1 media control;
five biological replicates;
BioLector and 20 ml fermentations
Analysis BioLector: pH, biomass, dissolved oxygen
Analysis 20 ml ferm.: targeted metabolite analysis;
free amino acid analysis

Influence of bifidobacteria



Fermentation setup: 8 defined + 1 media control;
five biological replicates;
BioLector and 20 ml fermentations
Analysis BioLector: pH, biomass, dissolved oxygen
Analysis 20 ml ferm.: targeted metabolite analysis;
free amino acid analysis